Playing With Numbers: Maven Toy Sales Analysis

1. Project Overview and Objective

This project focuses on analysing retail sales data using Power BI to understand sales performance, product profitability, and regional business trends through interactive visualizations.

The objective is to develop a data-driven dashboard that tracks key performance indicators (KPIs), compares product categories and regions, and identifies opportunities to improve sales and profitability.

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2. Data Sources

- **Source Description and Timeline:** Retail sales dataset covering the period 2020–2025, containing information on orders, products, sales, profit, customer segments, and geographical regions. (https://mavenanalytics.io/data-playground/mexico-toy-sales)
- **Domain**: Retail / Commerce / Sales Analytics.

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3. Problem Statement

- To analyse overall sales and profit performance using key metrics such as Total Sales, Total Profit, Order Count, and Profit Margin.
- To compare sales and profitability across different product categories to identify the highest-performing and least-profitable categories.
- To evaluate monthly sales trends and identify changes in sales performance over time.
- To compare regional sales performance and identify geographical areas contributing the most to total revenue.
- To analyse customer segment performance to understand which segments contribute most to sales and profitability.

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4. Attribute (Column /Features) Details:

Attribute Name	Data Type	Description
Store_ID	Text	Unique identifier for each Store
Store_Name	Text	Name of the Store
Store_City	Text	City in which the store is located
Store_Location	Text	Location category or area of the store
Store_Open_Date	Date	Date on which the store commenced operations.

Sale_ID	Text	Unique identifier assigned to each Sale
Date	Date	Date on which the sales transaction occurred
Product_ID	Whole Number	Unique identifier assigned to each product sold
Units	Whole Number	Number of units sold in the transaction
Product_Category	Text	Category to which the sold product belongs
Product_Cost	Decimal Number	Cost price of one unit of the product
Product_Price	Decimal Number	Selling price of one unit of the product
Revenue	Decimal Number	Total sales revenue generated from the transaction
Total Cost	Decimal Number	Total cost incurred for the products sold
Profit	Decimal Number	Profit earned from the transaction
Product_Name	Text	Name of the product available for sale
Stock_On_Hand	Whole Number	Current quantity of the product available in the store's inventory

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5. Tools & Technologies

- **Excel:** Used for data cleaning, data transformation, data validation, and initial data analysis using Pivot Tables.
- **Power Query:** Used for data preprocessing, handling missing values, checking duplicates, and preparing data for analysis.
- **Power BI:** Used for data modelling, creating DAX measures, KPI calculations, interactive visualizations, and dashboard development.

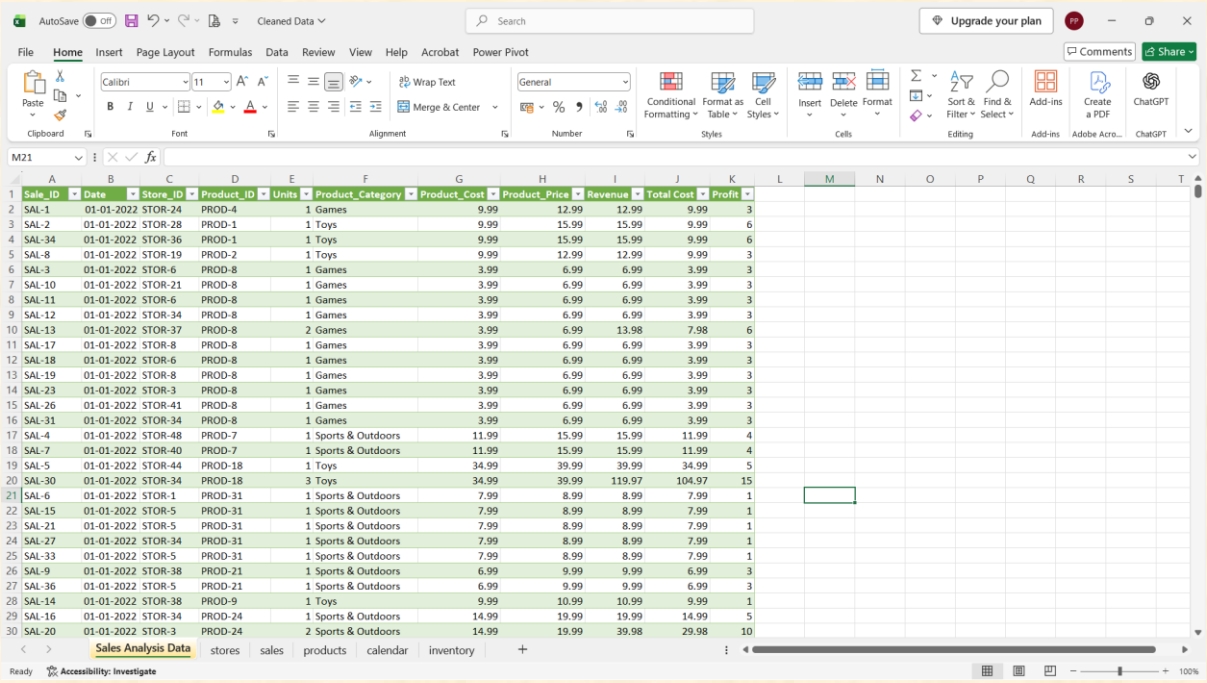
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6. Data Pre-Processing (Excel / Power Query)

Tasks Performed:

- **Data Inspection:** Reviewed the sales dataset and examined individual sheets to understand the structure, columns, and data quality.
- **Data Cleaning:** Checked for duplicate Order IDs and missing values, particularly in the Actual_Delivery_Date column.
- **Data Validation:** Verified delivery dates against order statuses and confirmed that delivered orders did not have missing actual delivery dates.
- **Data Transformation:** Used Power Query to clean and prepare the sales data for further analysis.

- **Data Preparation:** Created and organized the required tables to facilitate data modelling and Power BI dashboard development.

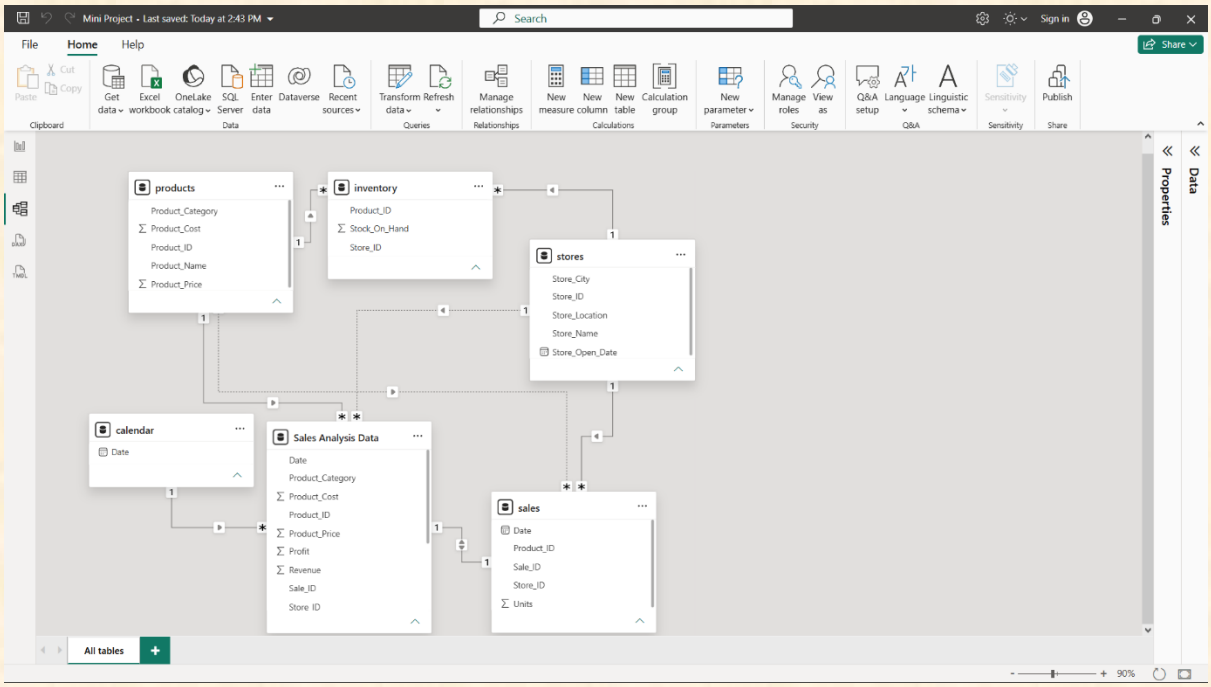


Sale_ID	Date	Store_ID	Product_ID	Units	Product_Category	Product_Cost	Product_Price	Revenue	Total Cost	Profit
SAL-1	01-01-2022	STOR-24	PROD-4	1	Games	9.99	12.99	12.99	9.99	3
SAL-2	01-01-2022	STOR-28	PROD-1	1	Toys	9.99	15.99	15.99	9.99	6
SAL-34	01-01-2022	STOR-36	PROD-1	1	Toys	9.99	15.99	15.99	9.99	6
SAL-8	01-01-2022	STOR-19	PROD-2	1	Toys	9.99	12.99	12.99	9.99	3
SAL-3	01-01-2022	STOR-6	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-10	01-01-2022	STOR-21	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-11	01-01-2022	STOR-6	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-12	01-01-2022	STOR-34	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-13	01-01-2022	STOR-37	PROD-8	2	Games	3.99	6.99	13.98	7.98	6
SAL-17	01-01-2022	STOR-8	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-18	01-01-2022	STOR-6	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-19	01-01-2022	STOR-8	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-23	01-01-2022	STOR-3	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-26	01-01-2022	STOR-41	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-31	01-01-2022	STOR-34	PROD-8	1	Games	3.99	6.99	6.99	3.99	3
SAL-4	01-01-2022	STOR-48	PROD-7	1	Sports & Outdoors	11.99	15.99	15.99	11.99	4
SAL-7	01-01-2022	STOR-40	PROD-7	1	Sports & Outdoors	11.99	15.99	15.99	11.99	4
SAL-5	01-01-2022	STOR-44	PROD-18	1	Toys	34.99	39.99	39.99	34.99	5
SAL-30	01-01-2022	STOR-34	PROD-18	3	Toys	34.99	39.99	119.97	104.97	15
SAL-6	01-01-2022	STOR-1	PROD-31	1	Sports & Outdoors	7.99	8.99	8.99	7.99	1
SAL-15	01-01-2022	STOR-5	PROD-31	1	Sports & Outdoors	7.99	8.99	8.99	7.99	1
SAL-21	01-01-2022	STOR-5	PROD-31	1	Sports & Outdoors	7.99	8.99	8.99	7.99	1
SAL-27	01-01-2022	STOR-34	PROD-31	1	Sports & Outdoors	7.99	8.99	8.99	7.99	1
SAL-33	01-01-2022	STOR-5	PROD-31	1	Sports & Outdoors	7.99	8.99	8.99	7.99	1
SAL-9	01-01-2022	STOR-38	PROD-21	1	Sports & Outdoors	6.99	9.99	9.99	6.99	3
SAL-36	01-01-2022	STOR-5	PROD-21	1	Sports & Outdoors	6.99	9.99	9.99	6.99	3
SAL-14	01-01-2022	STOR-38	PROD-9	1	Toys	9.99	10.99	10.99	9.99	1
SAL-16	01-01-2022	STOR-34	PROD-24	1	Sports & Outdoors	14.99	19.99	19.99	14.99	5
SAL-20	01-01-2022	STOR-3	PROD-24	2	Sports & Outdoors	14.99	19.99	39.98	29.98	10

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7. Data Modelling and DAX (Power BI)

- **Data Model:** Loaded six tables – Sales Analysis Data, Products, Stores, Inventory, Calendar, and Data_Dictionary – into Power BI and established relationships between the relevant tables using common keys.



- **Relationship Management:** Configured table relationships to enable accurate analysis of sales, products, stores, and time-based performance.
- **DAX Measures:** Created measures to calculate Total Revenue, Total Product Cost, Total Profit, and Profit Margin.
- **KPI Calculations:** Used DAX measures to generate key performance indicators for monitoring overall sales and profitability.
- **Data Formatting:** Applied appropriate number and percentage formats to KPI measures for clear dashboard presentation.
- **Dax Formula Used:**
  - Total Revenue = SUM('Sales Analysis Data'[Revenue])
  - Total Profit = SUM('Sales Analysis Data'[Profit])
  - Total Units Sold = SUM('Sales Analysis Data'[Units])
  - Profit Margin = DIVIDE([Total Profit],[Total Revenue],0)

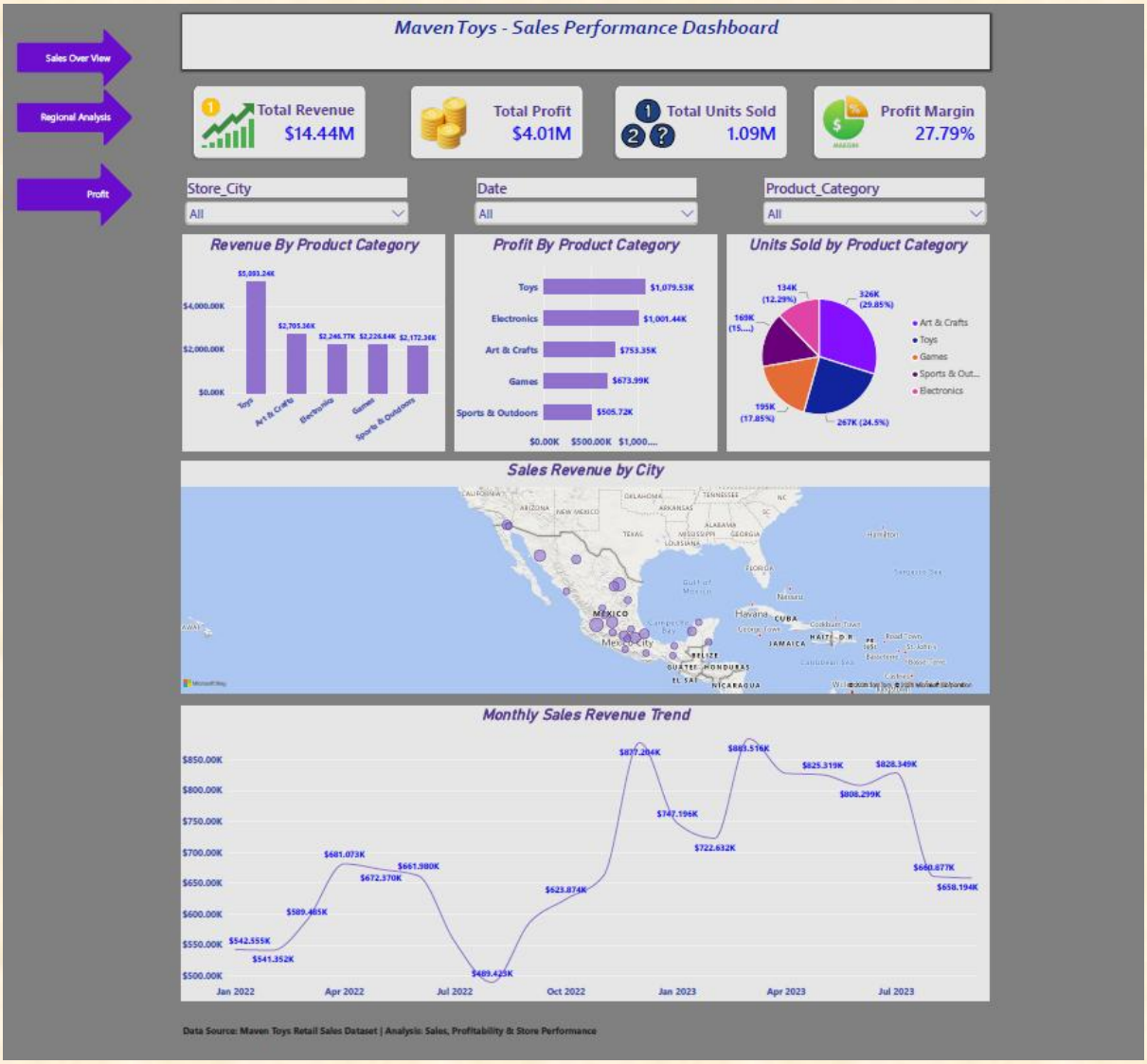
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8. Analysis and Visualizations (Power BI)

Dashboard Features:

- **KPI Cards:** Displayed Total Revenue, Total Product Cost, Total Profit, and Profit Margin to monitor overall business performance.
- **Sales Performance Analysis:** Used visualizations to analyse sales distribution across different product categories and geographical locations.
- **Monthly Sales Trend:** Created a line chart with drill-down functionality to explore sales performance by year, quarter, month, and day.
- **Product Category Analysis:** Implemented drill-down from product categories to individual product names to examine detailed sales performance.
- **Profitability Analysis:** Used charts and KPI indicators to evaluate product costs, total profit, and profit margins.
- **Regional Analysis:** Visualized sales performance across cities and regions to identify high-performing locations.
- **Interactive Filters and Slicers:** Added slicers to filter dashboard information and dynamically explore different sales dimensions.
- **Bookmarks and Navigation:** Created three bookmarks – Sales Overview, Profitability Analysis, and Regional Analysis – to provide different analytical views within a single-page dashboard.
- **Dashboard Design:** Developed a consolidated interactive dashboard with appropriate titles, data labels, charts, and navigation elements for clear presentation of business insights.

Dashboard Image



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9. Insights & Conclusions

- **Key Findings:**

  - **Sales Performance:** Evaluated overall revenue and sales performance using KPI indicators to understand business performance.
  - **Product Performance:** Analysed product categories and individual products to identify differences in sales contribution and profitability.
  - **Regional Performance:** Compared sales across different cities and stores to understand geographical sales distribution.
  - **Monthly Trends:** Examined monthly sales patterns to identify changes in revenue over different time periods.

- **Profitability:** Evaluated total profit, product costs, and profit margins to understand the profitability of the business.

- **Analysis insights:**
  - **Descriptive Analysis:**
    - ✓ Analysed overall sales performance using Total Revenue, Total Profit, Total Product Cost, and Profit Margin.
    - ✓ Summarised sales distribution across product categories, cities, and stores.
    - ✓ Identified monthly sales patterns through trend visualizations.
  - **Diagnostic Analysis:**
    - ✓ Compared revenue and profitability across different product categories to understand variations in performance.
    - ✓ Examined regional and store-level sales differences to identify factors contributing to higher or lower sales.
    - ✓ Evaluated product costs and profit margins to understand differences in profitability.
  - **Predictive Analysis:**
    - ✓ Used historical monthly sales trends to observe patterns and changes in revenue over time.
    - ✓ Identified recurring sales patterns that may help anticipate future sales performance.
    - ✓ Used historical performance as a reference for future sales planning.
  - **Prescriptive Analysis:**
    - ✓ Identified opportunities to improve profitability by focusing on products with better profit margins.
    - ✓ Recommended reviewing low-performing categories and regions to improve sales performance.
    - ✓ Suggested using sales trends and regional performance to support inventory planning and business decisions.

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10. Conclusions

This project transforms retail sales data into a structured analytical solution that goes beyond conventional sales reporting. By integrating data preparation, analytical modelling, and interactive visualization, the dashboard connects revenue, cost, and profitability across products, stores, and time periods. It provides a unified platform to examine business

performance, uncover patterns, and identify areas that require strategic attention. The project demonstrates how data analytics can bridge the gap between raw information and actionable business intelligence, enabling more informed, measurable, and data-driven decisions.